

Solution Requirements

Date	01 July 2026
Team ID	SWTID-2026-5245
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (farmers, researchers) sign up and log in using email/username and password. Passwords are hashed and stored securely; session tokens are used to keep the user logged in.	Medium
2	Authorization levels	Two roles are supported: Farmer (can enter soil/climate data and view crop recommendations) and Researcher/Admin (can view aggregated data, analytics dashboards, and manage crop datasets).	Medium
3	External interfaces	The system connects to a weather data API to fetch real-time temperature, humidity, and rainfall for the farmer's location, and uses the trained ML model (scikit-learn) served through a Flask/Streamlit backend.	High
4	Transactions processing	Each recommendation request (N, P, K, temperature, humidity, pH, rainfall) is validated, passed to the ML model, and the predicted crop	High

		with confidence score is returned and logged to the database in real time.	
5	Reporting	The system generates a crop recommendation report for farmers (recommended crop, resource-efficiency tips) and an analytics dashboard for researchers showing crop-environment trends, with CSV export of historical predictions.	Medium
6	Business rules, etc.	A recommendation is only generated once all seven required inputs (N, P, K, temperature, humidity, pH, rainfall) are provided and pass range validation (e.g., pH between 0-14, non-negative nutrient values).	High
7	Compliance to laws or regulations	User data (name, contact, farm location) is stored with basic data-protection practices; no sensitive personal or financial data is collected, minimizing regulatory exposure. Data usage follows institutional academic-project guidelines.	Low
8	Other	The system supports both single-crop suitability checks (Scenario 2) and full crop recommendation (Scenario 1) from the same input form, and allows model retraining as new labeled data becomes available.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The crop recommendation should be generated and displayed quickly after the farmer submits soil and climate data.	Prediction returned in under 2 seconds
2	Scalability	The system should handle increasing numbers of farmers, researchers, and prediction requests as adoption grows, without major redesign.	Supports up to 1,000 concurrent users
3	Security & Data Privacy	User credentials and farm data are protected using hashed passwords, HTTPS, and access control based on user role.	Passwords hashed (bcrypt); data transmitted over HTTPS
4	Reliability &	The deployed application should remain accessible with minimal downtime, since	99% uptime per month

	Availability	farmers may rely on it during sowing season.	
5	Usability & Accessibility	The input form and results screen should be simple and clear enough for farmers with limited technical literacy to use on mobile or desktop.	Mobile-responsive UI; minimal-click form submission
6	Other	The system should be maintainable and portable, allowing the ML model to be retrained/updated and the app redeployed on a new host with minimal changes.	Model retraining and redeployment completed within 1 day